# Atomic GPS Base

SKU:A134

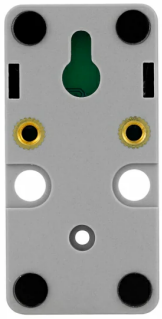













## Description

**Atomic GPS Base** is a GPS positioning module based on the Atom series host. The positioning and navigation chip is M8030-KT, with built-in FLASH and a coin cell battery, supporting power-off saving of user configurations. It uses the NMEA-0183 protocol for output and supports multiple satellite systems including GPS, GLONASS, GALILEO, BDS, SBAS, and QZSS. With 72 search channels, it can be used in applications such as geographic coordinate viewing, bus stop announcements, vehicle and ship navigation, and trajectory tracking. Additionally, there is a built-in TF (microSD) card slot below the GPS module, allowing you to read GPS or other file data, such as exporting GPS data in a specific format to view motion trajectories in mapping software, or using it as a regular card reader for file reading and writing.

## Features

- Compatible with Atom-Lite/Atom-Matrix/AtomS3/AtomS3-Lite
- High capture sensitivity
- Supports single-system positioning or multi-system combined positioning with BDS / GPS / GLONASS / GALILEO / SBAS / QZSS
- Built-in spring-loaded TF (microSD) card slot
- Low power consumption design
- UART parameter settings:
  - Baud rate (9600bps)
  - Start bit (1 bit)
  - Stop bit (1 bit)

- Parity bit (none)

## Includes

- 1 x Atomic GPS Base

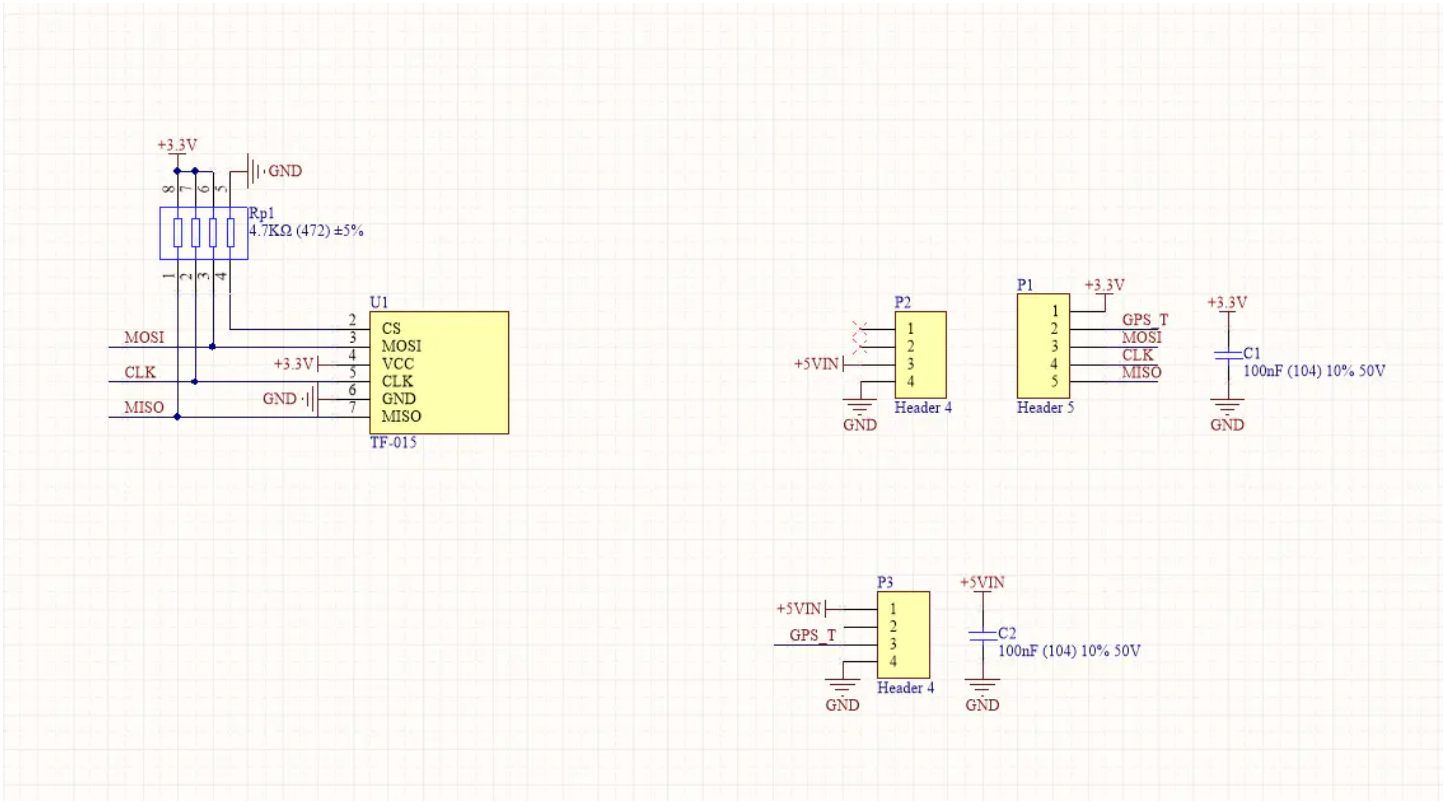
## Applications

- Vehicle and ship positioning and navigation
- Trajectory recording
- File reading and writing

## Specifications

| Specification | Parameter |
|---|---|
| Frequency | GPS L1, GLONASS L1, BDS B1, GALILEO E1, SBAS L1, QZSS L1 |
| Accuracy | Horizontal: 2m, Speed: 0.1m/s, Time: 1us |
| Number of Channels | 72 search channels |
| Update Rate | 1-10Hz |
| Maximum Speed | 515 m/s |
| Maximum Acceleration | < 4g |
| Sensitivity | Tracking: -167dBm, Acquisition: -160dBm, Cold Start: -148dBm, Hot Start: -156dBm |
| Start Time | Cold Start: 26 seconds, Warm Start: 25 seconds, Hot Start: 1 second |
| Baud Rate | 9600bps |
| Output Protocol | NMEA-0183 |
| NMEA Sentences | RMC, VTG, GGA, GSA, GSV, GLL |
| Indicators | TX: Blue light flashes when powered on, indicating data output; PPS: Flashes after 3D positioning, does not light up when not positioned |
| Operating Temperature | 0°C - 40°C |
| Product Size | 48.0 x 24.0 x 17.5mm |
| Package Size | 136 x 92 x 20mm |
| Product Weight | 14.4g |
| Gross Weight | 18.1g |

## Schematics

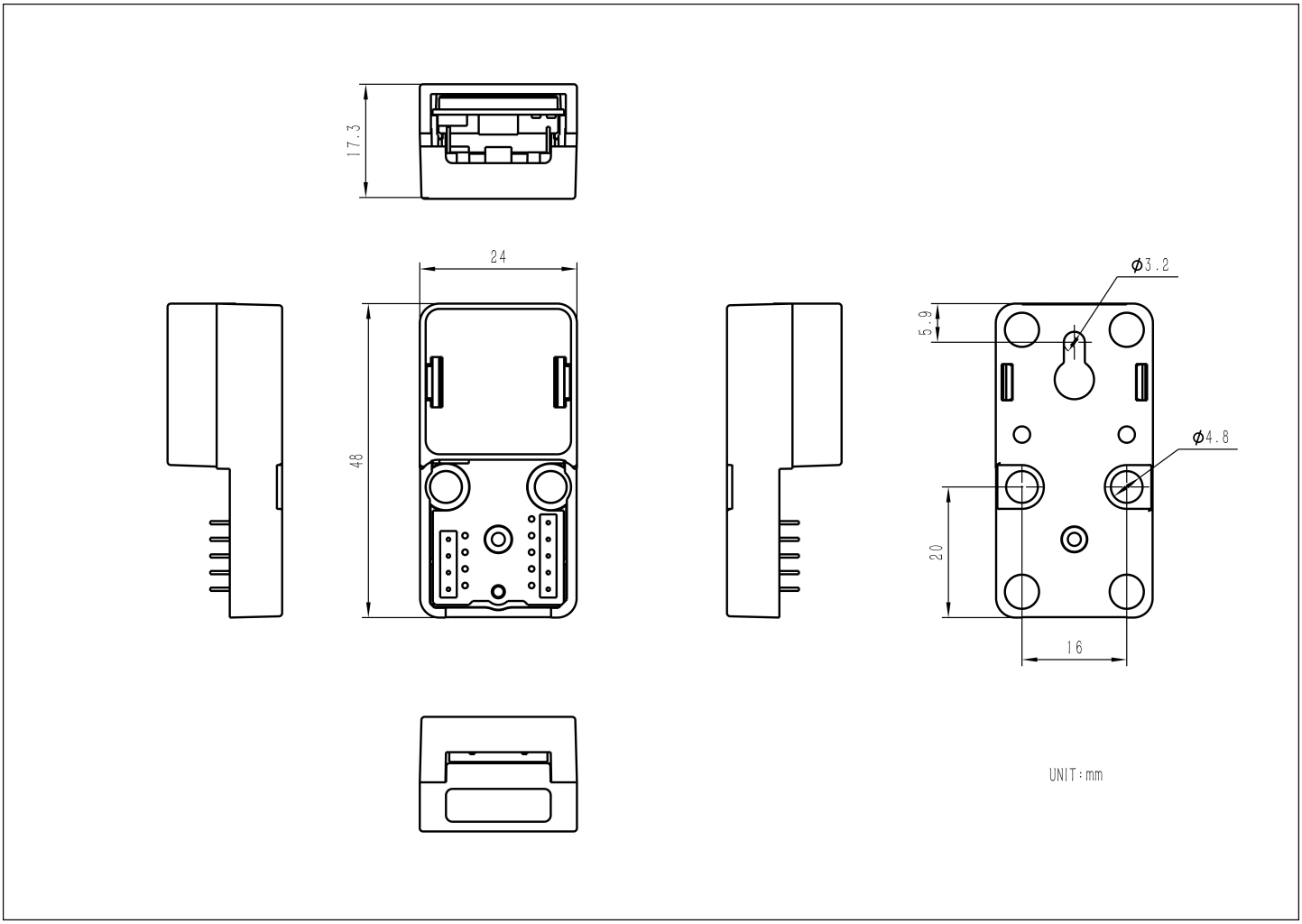



## PinMap

| PIN | LEFT | RIGHT | PIN |
|---|---|---|---|
| | | 1 | 3V3 |
| | 2 | 3 | UART_TX |
| | 4 | 5 | SPI_MOSI |
| 5V | 6 | 7 | SPI_CLK |
| GND | 8 | 9 | SPI_MISO |

## Model Size

- Atomic GPS Base Model Size PDF




# Softwares

## Arduino

- Atomic GPS Base Arduino Quick Start
- Atomic GPS Base microSD Example - with Atom
- Atomic GPS Base Location Example - with AtomS3

## UiFlow1

- Atomic GPS Base UiFlow1 Docs

## UiFlow2

- Atomic GPS Base UiFlow2 Docs

## Protocol

- CASIC multi-mode satellite navigation receiver protocol specification
- ATOMGPS method of changing the navigation system

## EasyLoader

| Easyloader | Download Link | Notes |
| --- | --- | --- |
| Atomic GPS Base Location Example Easyloader | download | / |

# Video

- Connecting to a mobile wireless serial port tool to view GPS information

AtomGPS.mp4